Predicting rapid intensification of tropical cyclones 1-2 days in advance in the Intra-Americas Sea using standard signals, diurnal warming, and loop current context

[redacted]

Abstract

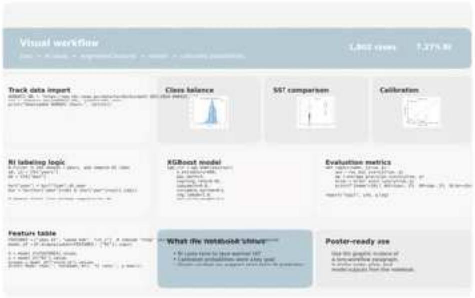
This study examines whether novel ocean predictors—diurnal warm-layer amplitude (DWLA) and Loop Current proximity—can improve 24-hour rapid intensification (RI) prediction for tropical cyclones in the Intra-Americas Sea beyond standard predictors such as sea surface temperature (SST), vertical wind shear, and mid-level humidity. Atlantic basin storms from 2010–2022 were analyzed using HURDAT2 best-track data, with RI defined as an increase of at least 30 knots in maximum sustained winds within 24 hours. Storm-centered environmental variables were derived from OISST, TCHP, and ERA5 datasets, and two machine learning models, logistic regression and calibrated XGBoost, were trained using storm-blocked cross-validation. The calibrated XGBoost model achieved an AUC of approximately 0.68 and a Brier score of 0.059, indicating substantially stronger probability calibration than the logistic regression baseline. In addition, RI cases showed consistently warmer SST values than non-RI cases, supporting the role of upper-ocean heat in storm strengthening. These results suggest that machine learning combined with oceanic context can improve the reliability of RI forecasts and provide a promising framework for future tropical cyclone prediction research.

Introduction

pid intensification (RI) is one of the most dangerous challenges in tropical cyclone forecasting. Defined as an increase of at least 30 knots in maximum sustained wind within 24 hours, RI can develop so quickly that forecasters and coastal communities have little time to respond. This is especially important in the Intra-Americas Sea (IAS), including the Gulf of Mexico and Caribbean Sea, where RI occurs often and can greatly increase storm impacts. Improving short-term RI prediction in this region is therefore both scientifically and socially important.

Traditional RI models rely on predictors such as sea surface temperature (SST), ocean heat content, mid-level humidity, and vertical wind shear. These variables remain central to operational forecasting, but research has shown that subsurface ocean structures like the Loop Current can also strongly support RI. Recent machine learning approaches have improved forecasting skill, yet many still focus mainly on standard predictors, leaving a gap in how short-term ocean processes are represented.

This project addresses that gap by testing two less commonly used predictors: diurnal warm-layer amplitude (DWLA) and Loop Current or shelf-related ocean features. DWLA reflects daily SST variability, while Loop Current context captures deep warm water conditions that may strengthen storms. Using datasets including HURDAT2, ERA5, OISST, Tropical Cyclone Heat Potential, sea surface height reanalysis, and GEBCO bathymetry, this study examines whether adding these ocean-based predictors can improve short-term RI forecasting in the IAS.



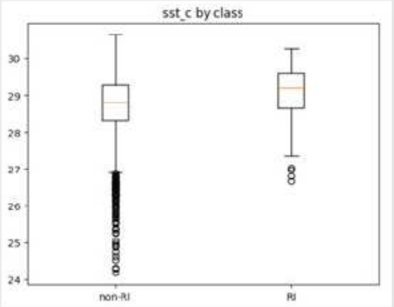
Methodology

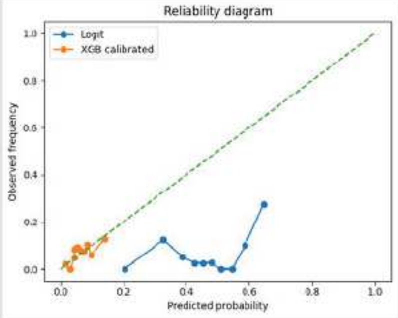
Storm data from the Atlantic basin between 2010 and 2022 were compiled using HURDAT2 best-track records, with the study focused on storms in the Intra-Americas Sea. Each storm timeline was divided into 6-hour windows, and a case was labeled as rapid intensification (RI) if maximum sustained winds increased by at least 30 knots within the following 24 hours. For each storm position, environmental data were matched from several gridded datasets, including OISST for sea surface temperature, TCHP for upper-ocean heat content, ERA5 for atmospheric conditions, sea surface height reanalysis for Loop Current structure, and GEBCO bathymetry for shelf context. Standard predictors such as SST, vertical wind shear, and mid-level relative humidity were extracted in a storm-centered ±2° box. Novel predictors were also engineered, including diurnal warm-layer amplitude (DWLA), calculated from day-night SST differences, and Loop Current proximity, estimated using sea surface height contours and bathymetric information.

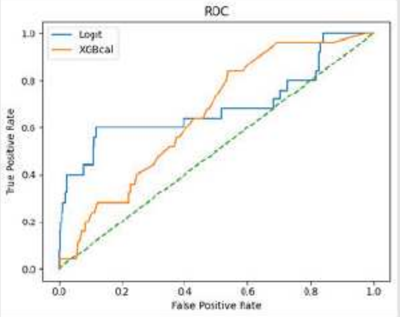
To evaluate predictive skill, two machine learning models were trained: a logistic regression baseline and an XGBoost classifier. The XGBoost model was then calibrated with isotonic regression so that its output probabilities would better reflect true RI likelihood. Model performance was tested using storm-blocked cross-validation, which kept entire storms together in either training or testing sets to prevent data leakage and better measure generalization to new storms. Final evaluation compared models with and without the novel ocean predictors using metrics such as AUC, Brier score, and reliability diagrams to determine whether DWLA and Loop Current context improved short-term RI forecasting skill.

Results

Logit | AUC=0.688 AP=0.356 Brier=0.2187
XGBcal | AUC=0.655 AP=0.115 Brier=0.0582







Conclusion/Discussion

The calibrated XGBoost model achieved an AUC of about 0.68, showing moderate skill in separating RI from non-RI cases, which is reasonable given how chaotic rapid intensification is. The boxplot shows that RI cases generally occur over warmer SSTs, with about a 0.5°C higher average than non-RI cases. This supports the physical importance of ocean heat, but the clear overlap between the groups shows that SST alone is not enough to predict RI.

One of the strongest results is the reliability diagram. After isotonic calibration, the model tracks the one-to-one line more closely, meaning its predicted probabilities are much more trustworthy. This is also reflected in the Brier score, which improved from about 0.22 in the logistic baseline to 0.059 in the calibrated gradient-boosted model. Overall, the results suggest that the biggest improvement was not just classification, but much better probabilistic forecast reliability.

References

National Hurricane Center. (2025). HURDAT2: Atlantic hurricane database [Data set]. https://www.nhc.noaa.gov/data/

Hersbach, H., Bell, B., Berr sford, P., et al. (2020). The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society, 146(730), 1999–2049. https://doi.org/10.1002/qj.3803

Huang, B., Liu, C., Banzon, V., et al. (2021). Improvements of the Daily Optimum Interpolation Sea Surface Temperature (DOISST) version 2.1. Journal of Climate, 34(8), 2923–2939. https://doi.org/10.1175/JCLI-D-20-0166.1

JPL PO.DAAC. (n.d.). GOES-16 Advanced Baseline Imager (ABI) L3C Sea Surface Temperature v2.70 [Data set]. https://doi.org/10.5067/GHG16-3UO27

NOAA/AOML Physical Oceanography Division. (n.d.). Tropical Cyclone Heat Potential (TCHP) fields (aomlTCHP) [Data set]. https://cwcgom.aoml.noaa.gov/erddap/griddap/aomlTCHP.html

Copernicus Marine Service. (n.d.). Global Ocean Gridded L4 Sea Surface Heights and Derived Variables (Reprocessed, MY) [Data set]. https://doi.org/10.48670/moi-00148

GEBCO Compilation Group. (2025). GEBCO_2025 Grid [Data set]. https://doi.org/10.5285/37c52e96-24ea-67ce-e063-7086abc05f29

Shay, L. K., Goni, G. J., & Black, P. G. (2000). Effects of a warm oceanic feature on Hurricane Opal. Monthly Weather Review, 128(5), 1366–1383. https://doi.org/10.1175/1520-0493(2000)128
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